

## Living Beings And Their Structure

A) Choose the best alternative.

1) The part of a cell which plays a role in inheritance is?

- a) ~~nucleus~~                      b) cytoplasm  
c) plasma membrane        d) mitochondria

2) Which of the following cell does not have a nucleus?

- a) white blood cell            b) ~~cytoplasm red blood cell~~  
c) nerve cell                    d) muscle cell

3) Plant cell wall is made up of?

- a) proteins                      b) fats  
c) ~~cellulose~~                    d) mineral salts

4) The aggregation of cells is known as a/an?

- a) organ                        b) ~~tissue~~  
c) system                      d) organelle

5) Organelle without a cell membrane is?

- a) ~~ribosome~~                    b) golgi apparatus  
c) chloroplast                d) Nucleus

6) Kitchen of the cell is:

- a) mitochondria              b) endoplasmic reticulum  
c) ~~chloroplast~~                d) golgi apparatus

7) The cell organelle that is known as the power house of the cell is called:

- a) ribosome                    b) ~~mitochondrion~~  
c) leucoplast                d) golgi bodies



B) Very short answer type questions.

1) What is cell?

→ Cell is the ~~smallest~~ structural and functional unit of life. ✓

2) Name the largest cell in human body.

→ The largest cell in human body is ~~ovum~~. ✓

3) Define multicellular organisms.

→ An organism that is composed of more than one cell is called multicellular organisms. ✓

4) What is the main function of chromoplast?

→ The main function of chromoplasts is to synthesize and store pigments that give bright colours (yellow, orange, red) to flowers and fruits to attract pollinators. ✓

5) Name the organelles that give energy to the cell.

→ Mitochondria ✓

6) Write an important function of mitochondria.

→ They perform cellular respiration to produce energy (ATP) from food molecules. ✓

7) What is the powerhouse of cell?

→ Mitochondrion is the powerhouse of the cell. ✓

8) Which cell organelle regulates all the activities of the cell?

→ The Nucleus regulates all cellular activities. ✓

C) Short answer questions

1) Write the differences between plant cell and animal cell.



| → Feature   | plant cell                           | Animal cell                                                           |
|-------------|--------------------------------------|-----------------------------------------------------------------------|
| • cell wall | Present (made of cellulose).         | Absent.                                                               |
| • plastids  | Present (Chloroplasts etc).          | Absent.                                                               |
| • vacuole   | One large central <del>vacuole</del> | Small and temporary <del>vacuoles</del><br><del>One large centr</del> |
| • shape     | Fixed and rectangular.               | Irregular and circular/oval.                                          |

2) Write the difference between chloroplast and chromoplast.

| → Features  | Chloroplast                             | chromoplast                                           |
|-------------|-----------------------------------------|-------------------------------------------------------|
| • colour    | Green in color (contains chlorophyll.). | various colors like red, orange, yellow.              |
| • Functions | Traps sunlight to manufacture.          | Attracts insects for pollination and fruit dispersal. |

3) Write two functions of cell membrane.

- 1) It protects the internal components of the cell from the external environment. ✓
- 2) It regulates the movement of substances in and out of the cell due to its selectively permeable nature. ✓

4) Where are the chromosomes found in a cell? Write their functions.

- Chromosomes are found inside the nucleus of a cell.  
Functions: They contain genes (DNA) that carry hereditary information and transfer parental characteristics to their offspring. ✓

5) Mitochondria is also called power house of the cell. Give reason.

- Mitochondria are called the powerhouses of the cell because they oxidize nutrients during respiration to release energy, which is stored and supplies in the form of ATP. ✓



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~~metabolic~~ molecules for various cellular metabolic activities.

6) Vacuoles are stores house of the cell. Give reason.

→ Because they store excess water, nutrients, enzymes, and metabolic waste materials until they can be used or excreted.

7) Cell wall is protective layer of plant cell. Give reason.

→ The cell wall is a rigid, tough outer layer that protects plant cells against mechanical stress, changes in temperature, high wind speed and osmotic pressure while maintaining the structural shape of the cell.

8) Chlorophyll helps for the manufacture of food in green plants. Explain with reason?

→ Chlorophyll is a green pigment capable of trapping solar energy. It uses this captured light energy to convert carbon dioxide and water into glucose (food) and oxygen through the process of photosynthesis.

9) Explain why chloroplasts are found only in plant cells?

→ Plants are autotrophs; they must manufacture their own food to survive. Chloroplasts contain chlorophyll required for photosynthesis. Animals are heterotrophs and depend on plants or other organisms for food, meaning they have no functional need for chloroplasts.



10) Draw a well labelled diagram of plant cell  
animals cell

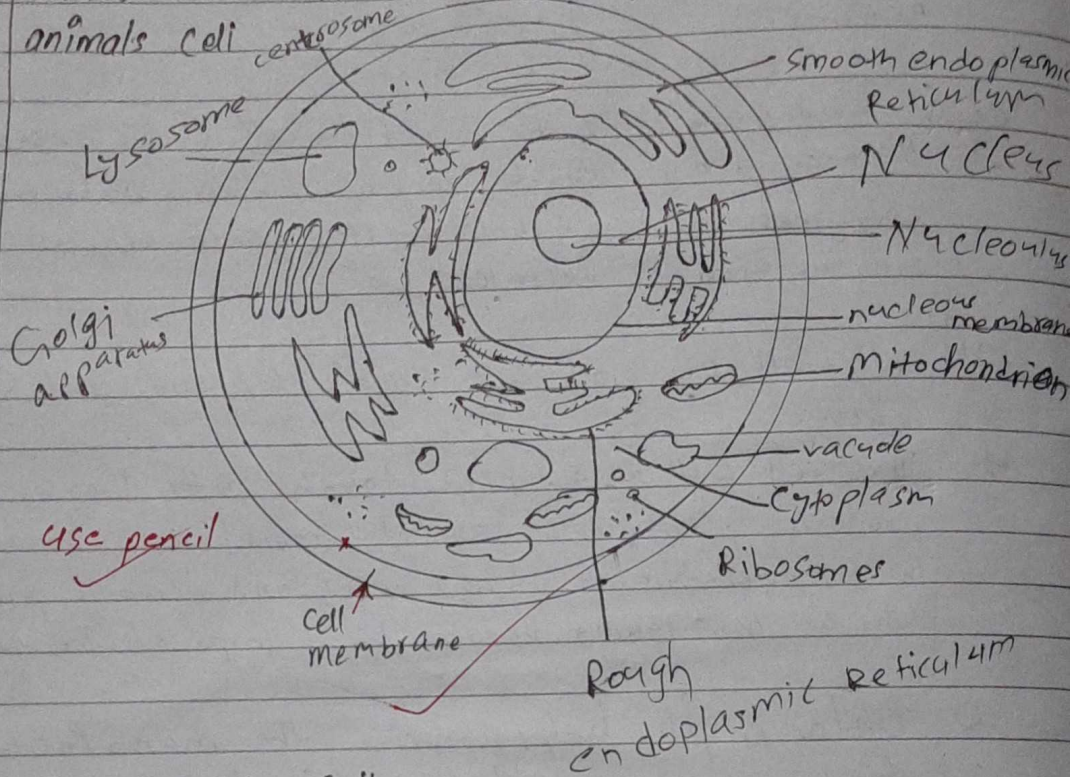


Fig. Animal Cell

11) Draw a well labelled diagram of plant cell.

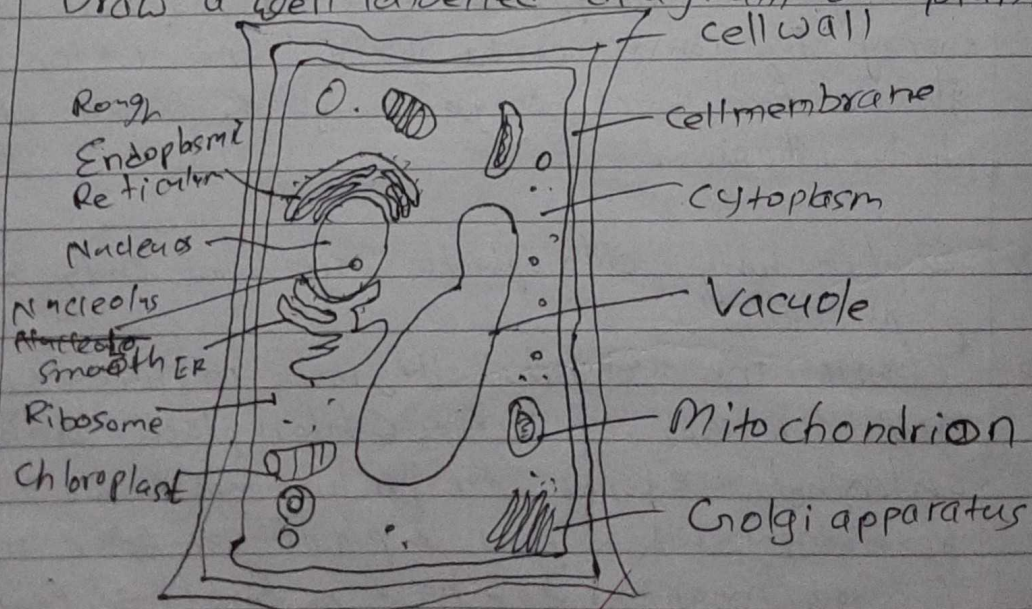


Fig. Plant Cell.



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12. plant cell possess large size vacuole than animal cells. Explain with reasons.

→ Plants cannot move to seek out water or support, so they rely heavily on internal fluid pressure. The large central vacuole stores large quantities of water and cellular sap to create turgor pressure, which keeps the plant cell rigid and helps the entire plant stand upright.

13) Why it is said that a cell without nucleus is without any future?

→ The nucleus acts as the "brain" of the cell, storing hereditary information (DNA) and sending instructions for protein synthesis, cell growth, and division. Without a nucleus, a cell cannot synthesize new proteins, repair its parts, or reproduce, meaning it will inevitably die shortly.

14) Why are lysosomes called "natural scavengers" and cellular house keepers?

→ Lysosomes contain powerful digestive enzymes that break down worn-out cell organelles, foreign bacteria, viruses, and waste debris inside the cell. By continuously clearing away this cellular waste, they keep the cell clean and functional.

D) Long answer questions:

1) If the organization of a cell is destroyed due to some physical or chemical influence, what will happen? Explain.

→ If the structural organization of a cell is catastrophically damaged, the cell loses its ability to carry out vital life processes, such as respiration, excretion,



nutrient absorption, and protein synthesis. When this breakdown happens, lysosomes often rupture, releasing their digestive enzymes into the cell body. These enzymes digest the cell's own internal components, resulting in autolysis (cellular self-destruction) and cell death.

2) Cells are the basic structural unit of living organisms. Justify.

→ Just as thousands of individual bricks are laid together to build the architecture of a house, every tissue, organ, and organ system in a living organism is built out of individual cells. No living organism can exist without at least one cell forming its structure. Since cells constitute the primary physical building blocks of life, they are justified as the basic structural units of living organisms.

3) When we study the different parts of a mustard plant, we find that root is colourless, leaf is green in colour and flower is yellow in colour. Explain these with suitable reasons.

→ This variation occurs due to the specialised distribution of different types of plastids across plant organs:

\* Leaves are green because they are packed with chloroplasts containing chlorophyll to harvest sunlight for photosynthesis.

\* Flowers are yellow because they contain chromoplasts filled with yellow pigments (like xanthophylls) meant to attract insects for pollination.

\* Roots are colourless because they grow underground away from sunlight, housing leucoplasts instead, which focus on storing food (starch and oils) rather than trapping light.



4) A, B, C, and D are basic parts of all the cells. The part A contains thread like structures called E which transfer the characters from parents to their offsprings 'into the cell' and 'out of the cell'. The part B is a double walled, spherical or rod like structures. The part C is selectively permeable and protects the ~~int~~ internal structure of the cell. The part of D is a transparent, jelly-like material. What could A, B, C, D and E be?

→ Based on the descriptive clues provided:

- A = Nucleus (contains hereditary material).
- B = Mitochondria (double-walled, rod-like structure).
- C = Cell membrane / plasma membrane (selectively permeable layer).
- D = Cytoplasm (transparent jelly-like base fluid).
- E = Chromosomes (thread-like structure carrying hereditary layer).

5) A cell has A as its control centre. A contains B that plays an important role in inheritance with the help of C. The cell also contains a jelly-like substance D. D contains various cell organelles like E that are essential for obtaining energy from food. The cell is surrounded by an F that is a living component. What could A, B, C, D, E and F be?

Also write a short note on C.

- 
- A = Nucleus
  - B = Chromosomes
  - C = Mitochondria
  - D = Cytoplasm
  - E = Cell Membrane (Plasma membrane)
  - F = Protoplasm (it consists both cytoplasm and a nucleus acting as the entire living mass).



6) X and Y are the two types of cells. The cell X have a well organized nucleus which is separated from the cytoplasm by the nuclear membrane. On the other hand, Cells Y do not have a real nucleus, their nuclear material is in direct contact with the cytoplasm.

(i) What type of cells are (i) X, and (ii) Y?

→ (i) X = Eukaryotic cell.

(ii) Y = Prokaryotic cell.

(ii) Give one example each of cells like (i) X, and (ii) Y.

→ (i) X = Onion cell (or Human cheek cell).

(ii) Y = Bacteria (E. coli) or Cyanobacteria (Blue-green alga).

7) The diagram given below shows a type of cell study it and answer the and answer the following questions?

(a) Identify the given cell.

→ Animal cell

(b) Label the part B and D.

→ B = Nucleus

D = Cytoplasm

(c) Write the function of part A and E.

→ Part A (Cell membrane): Regulates the entry and exit of molecules.

Part E (Mitochondria): Produces cellular energy (ATP) through respiration.

(d) How does it is important for living organism?

→ It functions as the foundational building unit of animal life, performing chemical processes that empower movement, growth, and tissue development.



8) The diagram given below shows a type of cell study it and answer the following questions?

a) Identify the given cell.

→ Plant cell.

b) Label the part A and D.

→ A = Cell wall

- D = Large central vacuole

c) Write the function of part C and E.

→ Part C (Chloroplast) = Absorbs sunlight to manufacture food via photosynthesis.

Part E (Nucleus) = Governs cell metabolism and coordinates cellular reproduction.

9) The parts A and B are present only in plant cell, they are not present in animal cells. The part A contains a green pigment called C whereas part B is made of a tough material D. The part A takes part in the food making process whereas part B gives shape and support to the plant cell.

10) Identify the parts A and B.

→ A = Chloroplast

B = Cell wall

b) Mention the name of C and D.

→ C = chlorophyll

D = cellulose

c) Write another importance of part B.

→ It prevents the plant cell from bursting (osmotic lysis) when it absorbs substantial volumes of water.



d) How it is different from animal cell. Write two points.

→ (i) Plant cells possess a rigid outer cellulose "cell wall", whereas animal cells only have a flexible cell membrane.

(ii) Plant cells contain "chloroplasts" to perform photosynthesis, which animal cells entirely lack.

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\* The ... End \*

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*27/3/17*